

SESSION 3 – Advanced Mobile Testing (APK, IPA) + Real Devices vs Emulators

1. APK Installation Testing

Ans:

- Downloaded a free Calculator APK file.
- Installed the application successfully using the Android Emulator.
- Installation completed without any errors or warnings.
- Application icon appeared in the app drawer after installation.

2. Screen Rotation Testing

Ans:

- Opened the Calculator application on the emulator.
- Rotated the device from portrait to landscape mode and back.
- UI adjusted automatically to both orientations.
- No major layout issues or screen distortions were observed.

3. Network Connectivity Testing

Ans:

- Disabled internet connectivity in the emulator.
- Attempted to access an online feature within the application.
- Application displayed a network connection error message.
- After restoring internet access, the feature worked normally.

4. Real Device vs Emulator Testing

Ans:

- Tested the application on both an Android emulator and a real Android device.
- The real device responded slightly faster during app launch.
- Push notifications appeared more reliably on the real device.
- UI rendering was similar, but touch interactions felt smoother on the physical device.

5. Push Notification Testing

Ans:

- Selected Instagram for notification testing.
- Verified notification permissions were enabled on both devices.
- Triggered notifications by sending messages and interactions.
- Notifications were received successfully on the real device.
- Emulator received notifications with a slight delay.
- Limitation: Emulators may not accurately simulate all notification services and background processes.

Result: Advanced mobile testing was completed successfully. The application performed as expected across both emulator and real-device environments, with only minor differences in speed and notification behavior.